

Two Patient Trauma



Section I: Scenario Demographics

Scenario Title:	Two for one MVC		
Date of Development:	01/03/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Martin Kuuskne
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	1) To appropriately triage simultaneous patients according to severity of injury 2) To downstream imaging and consultation
CRM Objectives:	To effectively communicate and lead a team through a complex situation where resources must be divided.
Medical Objectives:	1) To recognize and decompress an acute traumatic pneumothorax 2) To recognize and manage shock in the trauma patient with initiation of blood products/MTP 3) To recognize and manage respiratory compromise in the setting of a trauma

Case Summary: Brief Summary of Case Progression and Major Events

A young male and a middle-aged female are brought to the ED after a T-bone MVC at an unknown speed. Both patients were drivers. The emergency team is expected to triage the patients accordingly and to split the team so that both patients are treated.

Patient A: The team is expected to recognize respiratory compromise secondary to pneumothorax. Needle decompression and tube thoracostomy should be administered. The patient will remain in respiratory compromise post-decompression and the team should consider intubation. If the pneumothorax is not recognized or treated, the patient will arrest. On secondary survey, the patient will complain of pelvic pain in addition to a positive eFAST evaluation. The team should activate the massive transfusion protocol (MTP) and activate the trauma/surgery team.

Patient B: The team is expected to recognize hypoglycemia in the context of a minor head injury. Immediate glucose replacement is required.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case

Before entering the room: You are working the day shift in a tertiary care emergency department with full surgical capabilities. EMS is en-route to the hospital with two patients, a 37-year-old male and a 65 year old female, who were both drivers of a t-bone MVC of unknown speed. The ambulances will arrive in 2 minutes.

Upon entering the room: Each patient will be accompanied with a paramedic who will give this information and will be available to stay if asked.

Patient A: "37 year old male, belted driver, he got t-boned on the driver's side. There was significant intrusion of his side door. We're not sure if there was a loss of consciousness, we put him on a non-rebreather and his SAT was around 92%, tachy at 105 with an OK BP around 110 systolic during the ride."

Patient B: "65 year old female, belted driver who t-boned the other car. The front of her car was totaled. Airbags were deployed and there was a brief loss of consciousness. We put on the collar ASAP. Vitals were stable en route but she was a bit confused during the ride. No vomiting."

B. Scenario Cast & Realism

Patients: For both patients	☒ Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates	Brief Description of Role
Two EMS attendants	To give the HPI and to remain to help if asked.

C. Required Monitors

☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment

☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☒ Tube Thoracostomy Kit
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage

Patient A: Clothing apparel to suit age (37 yo M). Contusion/bruise to left axilla. Board & collar.

Patient B: Clothing apparel to suit age (65 yo F). Forehead abrasions. Board & collar.

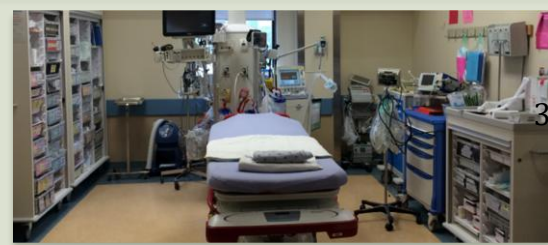
****Note:** Patient B could be played by a real person if access to two computerized mannequins is limited.

F. Approximate Timing

Set-Up:	5 min	Scenario:	12 min	Debriefing:	15 min
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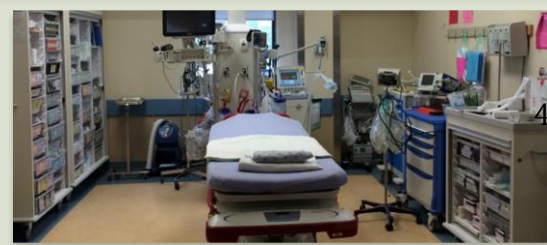
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Section V: Patient Data and Baseline State - Patient A

A. Patient Profile and History			
Patient Name: Chad		Age: 37	Weight: 85 kg
Gender: ☒ M ☐ F		Code Status: Unknown	
Chief Complaint: Severe diffuse pain and shortness of breath.			
History of Presenting Illness: Patient was driving at unknown speed and was making a left turn when he was T-boned by another car. The driver was belted, airbags were not deployed and there was a significant intrusion on driver's side door. Ambulance was called by bystanders.			
Past Medical History:	Nil	Medications:	Nil
Allergies: Nil			
Social History: unknown			
Family History: unknown			
Review of Systems:	Patient mutters about pain but unable to get focused review of systems from patient.		
B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display	☒ Monitor On, no data displayed		☐ Monitor on Standard Display
HR: 110/min	BP: 98/60	RR: 25 /min	O ₂ SAT: 87 %
Rhythm: sinus	T: 36.5 °C	Glucose: 6.0 mmol/L	GCS: 14 (E4 V4 M5)
General Status: Appears unwell, increased work of breathing and in pain			
CNS:	Muttering about pain, alert, moves all limbs to command		
HEENT:	Normal		
CVS:	Tachycardia, strong palpable pulses in all extremities		
RESP:	Minimal air entry of left side. No pooling of secretions. Airway patent.		
ABDO:	Normal		
GU:	Normal		
MSK:	Unstable pelvis. Tender with compression.	SKIN:	Left axillary area has contusion/bruise

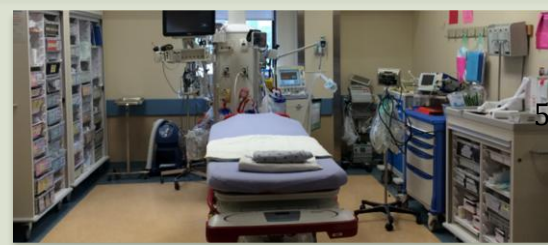
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Section VI: Scenario Progression –Patient A

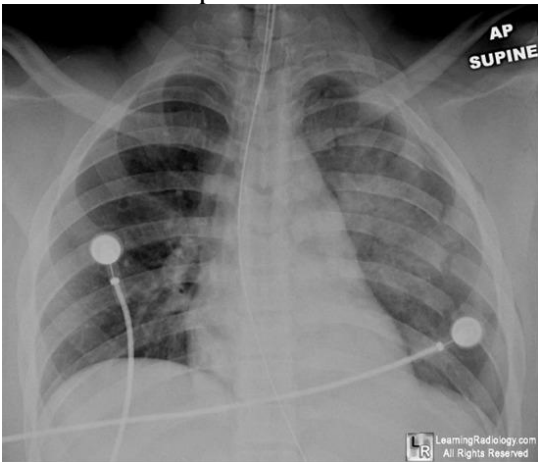
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tach HR: 110/min BP: 98 / 60 RR: 25/min O ₂ SAT: 87 % T: 36.5 °C	GCS – 14 (E4 V6 M6) – Patient moving all 4 limbs to command. Muttering he has pain. Decreased air entry on left side, flail segment.	<u>Learner Actions</u> ☐ Monitor, full vitals ☐ Supplemental O2 ☐ Establish IV access ☐ Take history from EMS ☐ Send trauma labs ☐ Needle decompression ± thoracostomy tube ☐ Verbalize and prepare for possible intubation ☐ CXR ☐ ± Perform eFAST ☐ Activate trauma team	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Needle decompression or finger thoracostomy → rush of air - eFAST → NO lung sliding on left <u>Triggers</u> <i>For progression to next state</i> - No decompression or thoracostomy by 3 min → 2. Worsening - Needle decompression → 3. Brief Improvement
2. Worsening HR: 120 bpm BP: 88/50 RR: 30 /min O ₂ SAT: 78 %	Increased respiratory distress.	<u>Learner Actions</u> ☐ Needle decompression ± thoracostomy tube ☐ Performs eFAST if not already done so ☐ Verbalize & prepare for intubation if not already done so	<u>Modifiers</u> - If no decompression in 2 min, RN to cue <u>Triggers</u> - Needle decompression → 3. Brief Improvement
3. Brief Improvement HR: 100 BP: 95/55 RR: 20 O ₂ SAT: 91%	Decreased respiratory distress but ongoing flail segment	<u>Learner Actions</u> ☐ Secondary Survey performed ☐ Prepare for intubation ☐ Thoracostomy tube ☐ Perform FAST ☐ Pelvic xray	<u>Modifiers</u> - FAST shows FF RUQ - PXR: open book # <u>Triggers</u> - 2 min into state → 4. Repeated Hypotension
4. Repeated Hypotension HR: 115 BP: 75/35 RR: 18 O ₂ SAT: 89%	Unchanged	<u>Learner Actions</u> ☐ Order blood products/initiate massive transfusion protocol ☐ Wrap pelvis ☐ Prepare for intubation ☐ Intubation	<u>Triggers</u> - 2 units PRBC given → 5. Improvement
5. Improvement HR: 95 BP: 115/75 RR: 12 (vent) O ₂ SAT: 92%	Unchanged	<u>Learner Actions</u> ☐ Call general surgery or ortho for OR or arrange for CT angio/interventional OR ☐ Intubation (if not yet) ☐ Positive Pressure Ventilation ☐ Post-intubation xray ☐ Initiate sedation	<u>Triggers</u> END CASE PRN

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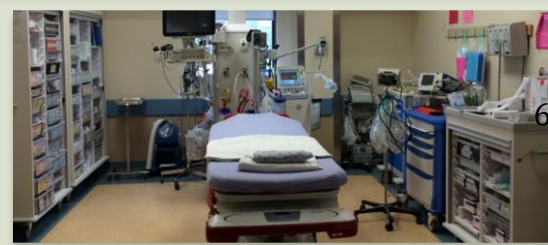


Section VII: Supporting Documents, Laboratory Results, & Multimedia – Patient A

Laboratory Results – Patient A					
Na: 130		K: 4.5		Cl: 105	
HCO ₃ : 21					
VBG	pH: 7.21	PCO ₂ : 30	PO ₂ : 40	HCO ₃ : 21	Lactate: 3.0
WBC: 14		Hg: 135			Plt: 250
INR: 1.1			PTT: 24.5		

Images (ECGs, CXRs, etc.)	
Patient A – Supine CXR  CXR source: http://learningradiology.com/archives2009/COW%20353-Flail%20Chest/caseoftheweek353page.htm	Patient A – Pelvic Xray  XR source: http://radiopaedia.org/articles/open-book-fracture
Ultrasound Video Files (if applicable)	
No lung sliding	FAST: Free fluid RUQ
No pericardial effusion	

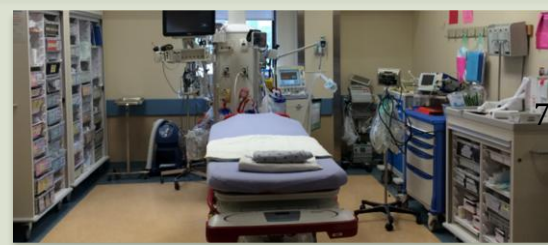
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Section VIII: Patient Data and Baseline State - Patient B

A. Patient Profile and History					
Patient Name: Michelle Bissidy		Age: 65		Weight: 75kg	
Gender: ☐ M ☒ F		Code Status: Unknown			
Chief Complaint: Head injury					
History of Presenting Illness: Patient drove into another car at an unknown speed. Patient was belted and airbags deployed. Brief loss of consciousness noted by EMS. Vitals stable and patient collared.					
Past Medical History:	Hypertension		Medications:	Hydrochlorothiazide	
	Dyslipidemia			Atorvastatin	
	Diabetes			Metformin	
				Insulin	
Allergies: nil					
Social History: unknown					
Review of Systems:	CNS:	Complaining of mild headache.			
	HEENT:	No complaints.			
	CVS:	No complaints.			
	RESP:	No complaints.			
	GI:	No complaints.			
	GU:	No complaints.			
	MSK:	No complaints.		INT:	No complaints.
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 90 /min		BP: 110/78		RR: 18/min	
Rhythm: sinus		T: 35.6°C		Glucose: 2.6 mmol/L	
				O ₂ SAT: 99%	
				GCS: 12 (E3 V4 M5)	
General Status: Appears well.					
CNS:	Seems confused. Somewhat agitated. Responding to pain, but not obeying commands.				
HEENT:	Normal				
CVS:	Palpable, strong, tachycardic, pulse in all extremities. No murmurs. Normal heart sounds				
RESP:	GAEB, no adventitious sounds				
ABDO:	Normal				
GU:	Normal				
MSK:	No extremity deformities		SKIN:	Forehead abrasions	

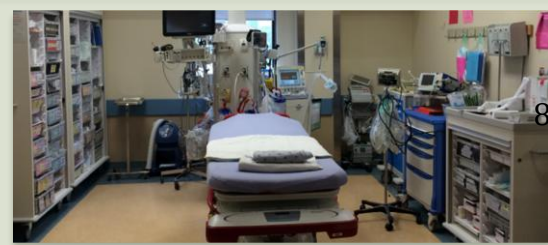
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Section IX: Scenario Progression –Patient B

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 90/min BP: 110/78 RR: 18/min O ₂ SAT: 99 % T: 35.6°C	GCS – 12 Seems confused. Responds to pain, but not obeying commands. Patient is collared.	<u>Learner Actions</u> - ☐ Monitor, full vitals - ☐ Establish IV access - ☐ Take history from EMS - ☐ Perform primary survey - ☐ Check cap sugar: 2.6 mmol/L - ☐ Replace glucose with D50 - ☐ Send trauma labs - ☐ ± eFAST - ☐ Progress to secondary survey - ☐ Determine need for further imaging (CT head, C-spine films)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - If patient not assessed → Patient becomes more agitated - If eFAST done → negative - If team decides needs imaging → radiology will ask who to image first <u>Triggers</u> <i>For progression to next state</i> - If cap sugar not checked by 5 min → 2. Decreased LOC - If sugar replaced, primary and secondary survey complete → 3. Normal GCS
2. Decreased LOC HR: 100 BP: 120/75 RR: 12	GCS to 7. E1 V2 M4 Patient essentially unresponsive	<u>Learner Actions</u> - ☐ Check cap sugar: 2.6 mmol/L - ☐ Replace glucose with D50 - ☐ Arrange for CT head - ☐ Consider other causes of aLOC	<u>Modifiers</u> - If preparing to intubate, RN to ask: “should we check a sugar first?” <u>Triggers</u> - Glucose replaced → 3. Normal GCS
3. Normal GCS HR: 90/min BP: 110/78 RR: 18/min O ₂ SAT: 99 %	GCS 15 Patient alert and oriented	<u>Learner Actions</u> - ☐ Send trauma labs - ☐ ± eFAST - ☐ Progress to secondary survey - ☐ Determine need for further imaging (CT head, C-spine films)	END CASE PRN

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Section X: Supporting Documents, Laboratory Results, & Multimedia – Patient A & B

Laboratory Results - Patient B					
Na: 140		K: 3.9		Cl: 108	
HCO ₃ : 22					
VBG	pH: 7.35	PCO ₂ : 40	PO ₂ :	HCO ₃ : 21	Lactate:1.1
WBC: 11		Hg: 110			Plt: 350
INR: 1.0			PTT: 26.5		

Images (ECGs, CXRs, etc.)	
Patient B – CXR  CXR source: http://radiopaedia.org/articles/normal-position-of-diaphragms-on-chest-radiography	

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Section XI: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	1) To appropriately triage simultaneous patients according to severity of injury 2) To downstream imaging and consultation
CRM Objectives:	To effectively communicate and lead a team through a complex situation where resources must be divided.
Medical Objectives:	1) To recognize and decompress an acute traumatic pneumothorax 2) To recognize and manage shock in the trauma patient with initiation of blood products/MTP 3) To recognize and manage respiratory compromise in the setting of a trauma
Sample Questions for Debriefing	
1. How was the decision made to split the team and resources? Do you think it was done well? 2. What was the team leader's leadership style? To oversee both traumas or to lead one case and assign a team member to lead the other case? 3. What difficulties are faced when using resources for a trauma involving multiple patients? 4. What is the appropriate treatment for flail chest? 5. How does the management of pelvic fractures change depending on patient stability?	
Key Moments	
Recognition that the team must be split to appropriately manage both patients.	
Triageing the patients.	
Recognition that Patient A continued to deteriorate despite the needle decompression/tube thoracostomy.	